Noether Sheet 5

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These were my solutions to the Noether Mentoring Scheme Sheet 5. Problem statements are abridged.

Problem 1

(a) Suppose that every positive integer is coloured red or blue. Show that there must be integers u, v, w and x (which need not all be different) of the same colour such that $u + v + w = x$. (This is called a *monochromatic* sum.)

(b) What is the largest N such that the integers $1, \ldots, N$ can be coloured red and blue without the sum $u + v + w = x$ having a monochromatic solution?

Solution

(a) Let us assume WLOG that 1 is blue. We can do this because if 1 was red then the arguement follows the same we just exchange the words red and blue. We argue by contradiction.

Note that if k and $3k$ are the same colour than we have a contradiction because $(k) + (k) + (k) = (3k)$ is a monochromatic sum.

Suppose that we have found a blue integer $k > 2$ such that $k + 1$ is red. Since k is blue, $3k$ must be red. Because there doesn't exist a monochromatic sum there cannot exist a red integer y such that

$$(k + 1) + (k + 1) + (y) = (3k) \iff y = k - 2$$

and so $k - 2$ must be blue. But now observe that

$$(k - 2) + (1) + (1) = (k)$$

is a monochromatic sum where every integer is blue, contradiction.

If we cannot find such a blue integer k then either the numbers $3, 4, 5, \ldots$ must be coloured

red, red, ... red, blue, blue, blue, ...

or such that every one of the numbers $3, 4, 5 \ldots$ is blue, or such that every one of those numbers is red. These are the only constructions, because any other construction would have a point at which k is blue and $k+1$ is red. In each case the colour of the numbers is eventually constant so clearly for large enough k we can find integers k, $3k$ that are the same colour, then we are done because of the reasoning presented in the second paragraph. □

(b) Again we can assume that 1 is blue. We look at two cases.

Case 1: If 2 is blue, then we have

1	2	3	4	5	6	7	8	9	10	11	12
B	B										

Now if k is blue you can make the monochromatic sums $(k) + (1) + (1)$, $(k) + (1) + (2)$, $(k) + (2) + (2)$. So if k is blue then $k + 2, k + 3$ and $k + 4$ must all be red. This gives

1	2	3	4	5	6	7	8	9	10	11	12
B	B	R	R	R	R						

Now note that k and $3k$ must be different colours. Also since 3 is red, if k is red then you can make the monochromatic sum $(k) + (3) + (3)$. So if k is red then $k + 6$ must be blue. This gives

1	2	3	4	5	6	7	8	9	10	11	12
B	B	R	R	R	R			B	B	B	B

But now since $(9) + (1) + (1) = (11)$ is a monochromatic sum with all numbers being blue, and we previously deduced that the number 11 must be blue, we have a contradiction. So in this case $N \leq 10$. In particular $N = 10$ is achievable with construction

1	2	3	4	5	6	7	8	9	10
B	B	R	R	R	R	R	R	B	B

which can be checked to work.

Case 2: If 2 is red, then we have

1	2	3	4	5	6	7	8	9
B	R							

Now since k and $3k$ cannot be the same colour, we have

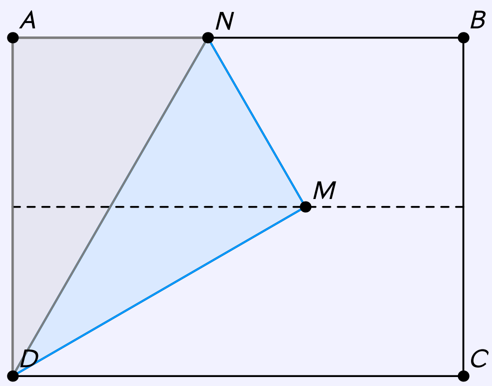
1	2	3	4	5	6	7	8	9
B	R	R			B			

But now since $(6)+(1)+(1) = (8)$ is a monochromatic sum with all numbers being blue, and $(3)+(3)+(2) = (8)$ is a monochromatic sum with all numbers being red, the number 8 can't be red or blue. So in this case $N \leq 7$.

Thus we have demonstrated that the largest such N is $N = 10$. ■

Problem 2

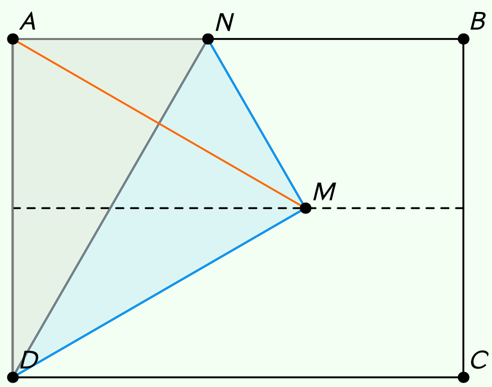
A rectangular piece of paper has a line drawn down its centre. One corner is folded such that the corner lies on the line, as shown in the diagram. (Here the top left corner is folded such that point A is sent to point M.)



What is angle $\angle ADN$?

Solution

We note that since the top left corner is folded, the segment AD is sent to DM, whilst AN is sent to NM. This means that $AD = DM$ and $AN = NM$. Now also since DN is common to both triangles $\triangle AND$ and $\triangle DMN$, we deduce that those two triangles are congruent by the SSS criterion. Now draw in the segment AM like so.



Now since A is just the reflection of D over that horizontal dashed line, and M lies on that dashed line it must be the case that $AM = DM$. So we can deduce that $\triangle AMD$ is in fact equilateral.

Moreover since $\triangle AND$ is congruent to $\triangle DMN$ we have $\angle ADN = \angle NDM$. So we deduce that

$$\angle ADN = \frac{1}{2}\angle ADM = \frac{1}{2}(60) = 30^\circ. \blacksquare$$

Problem 3

A domino in this question covers exactly two adjacent squares of a standard 8×8 chessboard.

(a) Can the entire chessboard be covered in dominoes, with every domino covering exactly two squares?

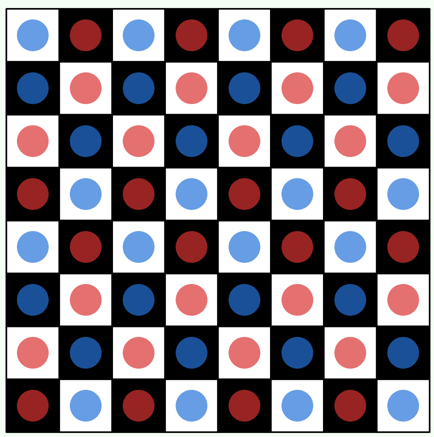
(b) One corner is removed from the chessboard. Can it now be covered in dominoes?

(c) Two opposite corners are removed. Can it now be covered in dominoes?

(d) If two squares are removed from the chessboard, under what condition can the board still be covered in dominoes?

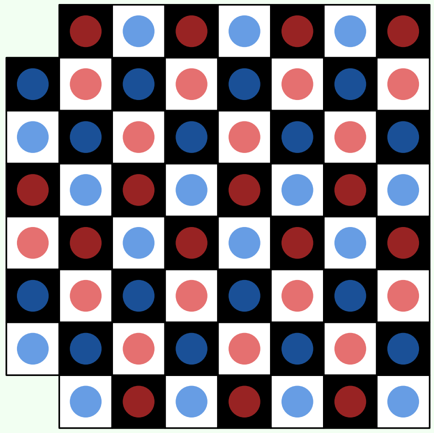
Solution

(a) The answer is yes. Here is one such way to do so.



(b) The answer is no. We say that two tiles are *neighbours* if they share a vertical or horizontal border. Each domino covers one pair of neighbours, that is exactly one white square and exactly one black square. So if the board can be completely covered in dominoes, then there has to be k white tiles and k black tiles. Since one corner is removed from the chess board there is a different number of white and black tiles on the board, which means that tiling it with dominoes is now impossible. □

(c) We assume that the question means two vertically/horizontally opposite corners. The tiling can't be done if we are speaking about diagonally opposite corners because of the same reasoning that was presented in (b). The answer to the first interpretation is yes. It suffices to show one construction for one pair of vertically opposite corners, because we can rotate the board to get the other 3 configurations of vertically/horizontally opposite corners being removed.



(d) Again because of the reasoning in (b), we must remove exactly 1 black square and exactly 1 white square. Now we will prove that this condition is necessary and sufficient.

It will be useful to us to introduce some notation to refer to which square we are talking about on a chess board. Let square (i, j) refer to the square on the i-th row from the top of the board and the j-th column from the left of the board. For example square $(1, 1)$ refers to the very top left square.

For some square (i, j) define the *weight* of that square to be the quantity $i + j \pmod 2$.

Claim 1: Two squares are the same colour, if and only if, they have the same weight.

Proof: Consider two squares (i, j) and (x, y). We note that if two squares are neighbours, then they have different colours, and also the two squares have different weights. We can travel from $(i, j) \to (x, y)$ by moving exactly $|x - i|$ squares vertically and $|y - j|$ squares horizontally. In total we have travelled $|x - i| + |y - j|$ squares.

- If we travel an even number of squares, then (i, j) and (x, y) are the same colour, because the coloured flipped an even number of times along our path, and $|x - i| + |y - j| \equiv (x - i) + (y - j) \equiv 0 \pmod 2 \implies x + y \equiv i + j \pmod 2$. Which means that (i, j) and (x, y) have the same weight.
- If we travel an odd number of squares, then (i, j) and (x, y) are different colours, since the colours flipped an odd number of times along our path, and $|x - i| + |y - j| \equiv (x - i) + (y - j) \equiv 1 \pmod 2 \implies x + y \not\equiv i + j \pmod 2$. Which means that (i, j) and (x, y) have different weights. $\square$

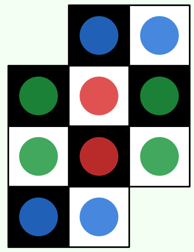
Claim 2: Let n, m be positive integers, with at least one of n, m even. Consider a $n \times m$ chessboard. We can always tile it with dominoes.

Proof: We assume WLOG that n is even. The construction is simple, for each of the m rows tile the row vertically with $n/2$ dominoes starting from the very top of the row. $\square$

Claim 3: Let n, m be positive integers of different parity. Consider a $n \times m$ chessboard. If the squares $(1, 1)$ and (n, m) are removed then, it is always possible to tile the resultant board with dominoes.

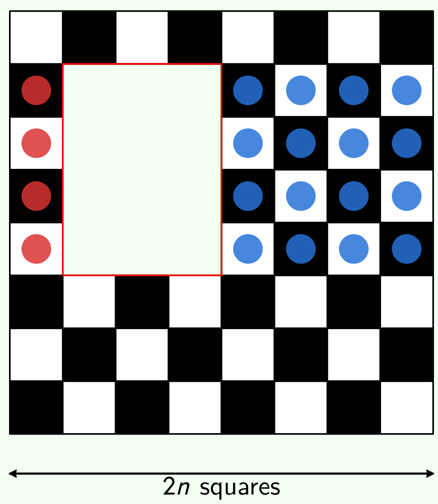
Proof: We can assume WLOG that n is even and m is odd. Since m is odd and we've removed the top left square and the bottom right square, there remains $m - 1$ squares on the top-most and bottom-most rows of the chess board. Since $m - 1$ is even we can tile these two rows horizontally with dominoes. Now we are just left with a $(n-2) \times m$ chessboard to tile, but this can be done due to claim 2. □

Here's an example illustrating claim 3 for the case $(n, m) = (4, 3)$.



Claim 4: Let a, b be postive integers of different parity. Let n be a positive integer satsifying $\max\{a, b\} \leq 2n$. Consider a $2n \times 2n$ chessboard with a $a \times b$ sub-chessboard removed from the board. The resultant shape can be tiled with dominoes.

Proof: WLOG let a be even, and b be odd. Consider squares on the same row as the $a \times b$ hole in the chessboard. These squares either fall on the left or on the right of the hole. All of the squares that fall on one particular side of the hole can be tiled, because these squares will make a sub-chessboard of size $a \times k$, recalling that a is even gives us the conclusion that this shape(s) can be tiled due to claim 2. Here is an illustration of the idea.



Finally to finish the proof we observe that all there is left to tile, is squares that lie directly above or below the $a \times b$ hole in the chessboard. To tile these shape(s) we can just make a rectangle with all of the squares that lie (above or below) the hole, such a rectangle will have exactly $2n$ squares on at least one of it's sides and so it is possible to tile it with dominoes again due to claim 2. □

Now let us return to the original problem. I will describe the high-level strategy. We will consider the "sub-chessboard" that is defined by these two removed squares. We will show that this sub-chessboard with it's two corner squares removed can be tiled using claim 3, then finally we will show that the remaining shape "a chessboard with a hole in it" can be tiled using claim 4.

Suppose we have removed two squares of distinct colours on an 8×8 chessboard, (x_1, y_1) and (x_2, y_2). Note that the two removed squares, define a sub-chessboard of dimensions $(|x_1 - x_2| + 1) \times (|y_1 - y_2| + 1)$. Due to claim 1 these two squares have different weights. We show that exactly one of the quantities

$$|x_1 - x_2| + 1, \quad |y_1 - y_2| + 1$$

must be even. It suffices to show that their sum is $\equiv 1 \pmod 2$. We have

$$\begin{aligned} &(|x_1 - x_2| + 1) + (|y_1 - y_2| + 1) \\ &\equiv (x_1 - x_2) + (y_1 - y_2) + 2 \\ &\equiv (x_1 - x_2) + (y_1 - y_2) \\ &\equiv (x_1 + y_1) - (x_2 + y_2) \\ &\equiv 1 \pmod 2 \end{aligned}$$

where the last line is because the two squares have different weights. So since this sub-chess board has dimension of the form $a \times b$, with $a, b \in \mathbb{N}$ and exactly one of a, b even, The $(|x_1 - x_2| + 1) \times (|y_1 - y_2| + 1)$ chessboard with two corners removed can be tiled with dominoes by applying claim 3. Finally the remaining untiled squares form an 8×8 chessboard with a $(|x_1 - x_2| + 1) \times (|y_1 - y_2| + 1)$ hole in it. Indeed exactly one of those numbers is even and

$$\max(|x_1 - x_2| + 1) = \max(|y_1 - y_2| + 1) = 8 - 1 + 1 = 8 \leq 8.$$

Thus this shape can indeed be tiled due to claim 4. ■

Comments: I am purposefully hand-waving with all these WLOG's, I'm not super sure on how to formally justify them tbh. Other than this I believe the problem is solved now, this one was particularly enjoyable. I have a feeling that my "chessboard with a hole in it" solution wasn't the intended one, so im curious to see what other (probably much simpler) approaches work.

Problem 4

(a) Suppose that $ABCDEF$ is a regular hexagon. If you connect the vertices AC, BD, CE, DF, EA and FB, then a smaller hexagon is formed.

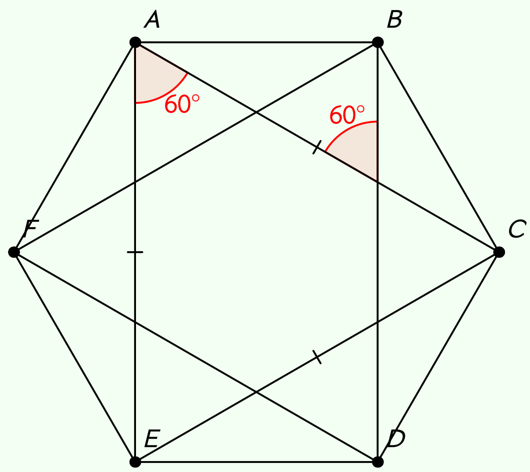
What is the ratio of the side lengths of the original hexagon to those of the smaller hexagon?

(b) Suppose that $ABCDE$ is a regular pentagon. If you connect vertices AC, BD, CE, DA and EB, then a smaller pentagon is formed.

What is the ratio of the side lengths of the original pentagon to those of the smaller pentagon?

Solution

(a) Here's a diagram. Let the regular hexagon have the 6-th roots of unity as vertices.



Since this is a regular hexagon, rotating the diagram by 120° anti-clockwise about the origin will send $C \to A$ and $A \to E$. Similarly rotating the diagram by 120° clockwise about the origin will send $E \to A$ and $C \to E$. Since the length of a segment is unchanged under rotation this allows us to conclude that $AC = CE = AE$.

Now since $\triangle ACE$ is equilateral, we have $\angle CAE = 60°$. Then note that $AE \parallel BD$, because they map to each other under a reflection in the line $x = 0$. Thus by alternate angles we deduce that $\angle AHB = 60°$ where $H = AC \cap BD$. Now since the hexagon $ABCDEF$ has side length 1, we deduce that

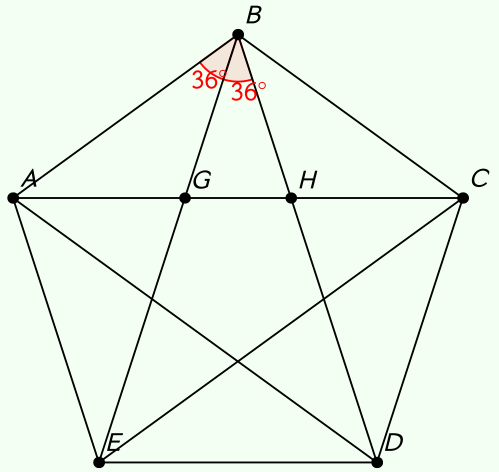
$$AH = \frac{1}{\sin 60°} = \frac{2}{\sqrt{3}}.$$

Now where $X = AC \cap BF$, we note that X lies on the midpoint of AH, because BF is just the reflection of AC over $x = 0$ and so the only point at which they intersect would be at $x = 0$. Thus

$$HX = \frac{1}{2}\frac{2}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

and so the desired ratio is $1 : \frac{1}{\sqrt{3}} = \sqrt{3} : 1$. $\square$

(b) Here's a diagram.



Firstly because $\angle EAB = 108°$ and $AB = AE$, $\triangle ABE$ is isoceles and so $\angle ABE = 36°$. By a similar arguement $\angle DBC = 36°$ which forces $\angle GBH = 36°$ since a regular pentagon have interior angles of $108°$.

Now if we let the regular pentagon have side length 1, we can use sine rule to find that

$$GB = \frac{\sin 36°}{\sin 108°} = \frac{\sin 36°}{\sin(180-108)°} = \frac{\sin 36°}{\sin 72°} = \frac{\sin 36°}{\cos 36°} = \tan 36°.$$

Then again by sine rule

$$GH = \frac{\sin 36°}{\sin 72°}GB = \frac{\sin 36°}{\cos 36°}GB = \tan^2 36°.$$

So the desired ratio is $1 : \tan^2 36°$.

Now we attempt to evaluate $\tan^2 36°$. We can use De Moivre's theorem as follows. Let $\theta = \pi/5$.

$$\begin{aligned}\sin 5\theta &= \text{Im } (e^{i\theta})^5 \\ &= \text{Im } \sum_{r=0}^{5} (i\sin\theta)^r \cos^{5-r}\theta \\ &= x(16x^4 - 20x^2 + 5)\end{aligned}$$

where $x = \sin\theta$. The roots of this polynomial correspond to

$$x = \sin\frac{k\pi}{5}, \quad (k = 0, 1, 2, 3, 4).$$

Clearly $k = 0$ corresponds to the lone factor of x so let's ignore it. If we solve that quartic in terms of x^2 we get

$$x^2 = \frac{5 \pm \sqrt{5}}{8}.$$

Since $\sin\theta$ is positive on the interval $(0, \pi)$, the 4 roots must be

$$\sqrt{\frac{5-\sqrt{5}}{8}}, \sqrt{\frac{5-\sqrt{5}}{8}}, \sqrt{\frac{5+\sqrt{5}}{8}}, \sqrt{\frac{5+\sqrt{5}}{8}}.$$

Now if you look at the graph of $\sin x$ you can see that $\sin \pi/5$ and $\sin 4\pi/5$ should be amongst the smallest of these values so

$$\sin\frac{\pi}{5} = \sqrt{\frac{5-\sqrt{5}}{8}}.$$

Now since both $\sin \pi/5$ and $\cos \pi/5$ are positive we can compute

$$\cos\frac{\pi}{5} = \sqrt{1-\left(\sqrt{\frac{5-\sqrt{5}}{8}}\right)^2} = \sqrt{\frac{3+\sqrt{5}}{8}}.$$

Now recalling that $\tan^2\theta = \sec^2\theta - 1$ we have

$$\tan^2\frac{\pi}{5} = \frac{8}{3+\sqrt{5}} - 1 = 5 - 2\sqrt{5}$$

and thus the desired ratio turns out to be $1 : (5-2\sqrt{5})$. ■

Problem 5

(a) Show that

$$\binom{n}{r} = \binom{n}{n-r}.$$

(b) Show that

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{n}^2 = \binom{2n}{n}.$$

Solution

(a) We note that

$$\binom{n}{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{(n-r)!r!} = \binom{n}{r}. \ \square$$

(b) This result is a direct application of Vandermonde's identity.

Theorem. Vandermonde's Identity. Let $q \leq \min\{m, n\}$ be positive integers. Then

$$\sum_{r=0}^{q} \binom{m}{r}\binom{n}{q-r} = \binom{m+n}{q}.$$

Proof: Consider

$$(1+x)^m(1+x)^n = (1+x)^{m+n}$$

$$\left[\sum_{r=0}^{m} \binom{m}{r} x^r\right]\left[\sum_{k=0}^{n} \binom{n}{k} x^k\right] = \sum_{t=0}^{m+n} \binom{m+n}{t} x^t$$

Now compare the coefficients of x^q,

$$\sum_{r=0}^{q} \binom{m}{r}\binom{n}{q-r} = \binom{m+n}{q}. \ \square$$

If we apply this theorem with $(q, m, n) = (n, n, n)$ we get

$$\sum_{r=0}^{n} \binom{n}{r}\binom{n}{n-r} = \binom{2n}{n}.$$

Now using the result from part (a) we can re-write this as

$$\sum_{r=0}^{n} \binom{n}{r}^2 = \binom{2n}{n}. \ \blacksquare$$

Problem 6

2021 has the property that it is formed by concatenating two consecutive numbers (20 and 21), and it is also the product of two numbers that differ by 4 (43×47).

Find all other 4 digit numbers with these properties.

Solution

I assume that when we say concatenating two consecutive numbers we write $\overline{x(x+1)}$. If writing $\overline{(x+1)x}$ is allowed the strategy detailed below doesn't seem to work.

Call a number *special* if it satisfies these properties. Indeed the number that is formed by concatenating two consecutive numbers $\overline{x(x+1)}$ can be expressed as $100x + (x+1) = 101x + 1$. The problem reduces to finding solutions to

$$101x + 1 = k(k+4)$$

for $k, x \in \mathbb{N}$. We can complete the square to find that

$$(k+2)^2 - 5 = 101x.$$

Subtract $20 \cdot 101$ from both sides, (see below for the motivation)

$$(k+2)^2 - 5 - 20 \cdot 101 = (k+2)^2 - 2025 = 101(x - 20)$$

Noting that $2025 = 45^2$ we complete the square,

$$(k-43)(k+47) = 101(x-20)$$

Since $k(k+4)$ must be a 4 digit number we require $k \le 99$. Now we note that 101 must divide one of $(k+47)(k-43)$.

- If $101 \mid (k+47)$ then using $1 \le k \le 99$ it's easy to show that only $k + 47 = 101 \implies k = 54$.
- If $101 \mid (k-43)$ then using $1 \le k \le 99$ it's easy to show that only $k - 43 = 0 \implies k = 43$.

From here it can be checked that $(k, x) = (54, 31), (43, 20)$ are the only solutions. So the only 4 digit special numbers are $3132,\ 2021$ (assuming the $\overline{x(x+1)}$ convention.) $\blacksquare$

Motivation: Once I reached the equation $(k+2)^2 - 5 = 101x$, I spent a long time trying to find solutions to $(k+2)^2 \equiv 5 \pmod{101}$, but I wasn't able to find anything here. Then after thinking a while about how I can use the fact that 101 is prime, I thought if I can factor the LHS I might be able to force unique values of k since the range for $1 \le k \le 99$ is fairly small. I looked for $a \in \mathbb{N}$ such that $5 + 101a$ is square so that I could factor the LHS with difference of two squares as desired. I wrote a quick python script and found the first two solutions $a = 20, 31$.

Re-reading over this I realise that finding a such that $101a + 5$ is square is essentially the same as my original problem of finding solutions to $(k+2)^2 \equiv 5 \pmod{101}$. Right now I can't see any better method other than trial and error to solve this congruence, or guessing to solve here.

Problem 7

Suppose that $p_1, p_2, \dots, p_n$ are all primes.

(a) Consider the quantity $p_1 p_2 \cdots p_n + 1$. Could this be divisible by any of the p_i ?

(b) Using this result prove that there are infinitely many primes.

(c) By considering the quantity $4p_1 p_2 \cdots p_n - 1$, show that there are infinitely many primes of the form $4k+3$.

Solution

(a) Notice that $p_1 p_2 \cdots p_n + 1 \equiv 1 \pmod{p_i}$. Using the Euclidean algorithm,

$$\gcd(p_1 p_2 \cdots p_n + 1, p_i) = \gcd(1, p_i) = 1.$$

So p_i never divides $p_1 p_2 \cdots p_n + 1$. $\square$

(b) If there are finitely many primes $p_1, p_2, \dots, p_n$ and $P = p_1 p_2 \cdots p_n + 1$ is coprime to all of the primes p_i, then due to the fundamental theorem of arithmetic P must have some prime factors, in particular it's prime factors aren't among $p_1, \dots, p_n$. But this contradicts the assumption that $p_1, \dots, p_n$ are all the primes. $\square$

(c) Suppose there are finitely many primes congruent to $3 \pmod 4$, say $q_1, q_2, \dots q_n$. Consider the quantity,

$$P = 4q_1 q_2 \cdots q_n - 1.$$

Note that for every q_i we have $P \equiv -1 \pmod{q_i}$, and so by the Euclidean algorithm,

$$\gcd(4q_1 q_2 \cdots q_n - 1, q_i) = \gcd(-1, q_i) = 1.$$

So for every q_i, we have $q_i \nmid P$. By the fundamental theorem of arithmetic P factors into a set of primes. Since P is odd, all of these primes are congruent to $\pm 1 \pmod 4$. However since P itself is congruent to $-1 \pmod 4$, P must have $2k+1, (k \in \mathbb{Z}_{\geq 0})$ pairwise distinct prime factors that are congruent to $-1 \pmod 4$. Moreover since $q_i \nmid P$, this means that there exists at least one new prime $q_{n+1} \equiv -1 \pmod 4$ which is not a part of the collection $q_1, \dots, q_n$, a contradiction. $\square$

Comments: I first saw the problem in part (c) at the Winchester maths summer camp in Year 12. At that time I had just started discovering contest math, so when the teacher (Dominic Rowland) tried explaining this proof to me I was lost and couldn't understand a thing. Good to know that now I can actually understand and explain the proof.

Here's a note explaining the Euclidean algorithm. Basically suppose we have $a, b \in \mathbb{Z}$ satisfying $a \equiv r \pmod b$. Then

$$\gcd(a, b) = \gcd(r, b).$$

This is because the gcd, g must be a divisor of b. Since we also have $g \mid a$, we require

$$\frac{a}{g} = \frac{kb + r}{g} = k\left(\frac{b}{g}\right) + \frac{r}{g}$$

to be an integer, which happens iff $g \mid r$.